



Haoran Yang

Target role: AI Infra / LLM Algorithm · Inference & training systems

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◆ EDUCATION

2026/09 – Present

The University of Hong Kong MSc in Artificial Intelligence

School of Computing and Data Science

Core courses: Deep Learning, Large Language Model, Nature Language Process

2019/09 – 2023/06

Sichuan University B.Eng. in Electronic Information Engineering

Core courses: C++ Programming, Data Structures, Operating Systems, Compiler Principles, Optimization Algorithms, Image Processing

◆ EXPERIENCE

2026/06 – 2026/08

Zhuoyu Technology Autonomous Driving System Engineer

Model post-training · Data mining

- Worked on the end-to-end post-training pipeline for autonomous-driving models, covering training data construction, fine-tuning and evaluation.
- Owned data mining and cleaning workflows, extracting high-value samples from raw driving data to support model iteration.

2024/07 – 2026/05

WeRide Infra System Engineer

Onboard Infra · ROS-like framework · System performance

- Developed and maintained onboard backend modules for autonomous driving; improved system stability to keep the self-driving stack reliable and efficient.
- Built a module monitoring system on the in-house ROS-like framework: data persistence, real-time reporting and distributed deployment.
- Developed and maintained the middleware between the autonomous-driving stack and chassis-by-wire, optimizing communication logic.
- On-call for the onboard system: handled on-site deployment and crashes, rapidly located and resolved failures.

2023/07 – 2024/07

Shanghai Yuliangsheng Technology Co., Ltd. Software Engineer

High-precision computing · Algorithm engineering · Performance

- Led production engineering of high-precision computing modules, translating algorithm logic into deployable code.
- Participated in modular refactoring and performance optimization of business components, delivering core features for the first time.

◆ PROJECTS

Personal Project

llm_go — End-to-End LLM Pipeline: Train → Fine-tune → Quantize → Serve

PyTorch · transformers · LoRA · llama.cpp · GGUF quantization · Go (Gin) · Docker

- Trained a small LLaMA-style model from scratch (~4M params, BPE tokenizer, GQA); wrote a custom GGUF exporter to get past the official tokenizer whitelist, then quantized to Q4_K_M with llama.cpp — about 1/4 of the F16 size.
- Built the quantization and measurement path: imatrix calibration plus size/perplexity comparison across schemes, unified measurement of time-to-first-token, prefill and decode throughput and server RSS, and the extra cost of KV-cache quantization.
- Extended two model-side experiment paths: LoRA instruction tuning (merged back into HF weights and reused through the same deploy pipeline), and draft + target speculative decoding with vocab-consistency checks, measuring acceptance rate and decode speedup.
- Built an inference gateway in Go (Gin) exposing an OpenAI-compatible API with streaming output; orchestrated llama-server and the gateway with Docker Compose.
- Implemented the visual orchestration UI (canvas and list views, preset templates): 11 pipeline nodes executed in topological order, logs streamed over SSE, and a Markdown report with a timing summary written for every stage.

Core Developer

WeTalk — Full-stack Distributed IM System in C++

Qt · gRPC · Boost · Asio · High-concurrency long connections

- Developed the DispatchServer gateway: HTTP APIs for login/registration, gRPC for service dispatch and authentication.
- Implemented asynchronous TCP long-connection communication based on Asio, with pooled io_context for higher concurrency.
- Built the Qt client and modular server design supporting distributed deployment, decoupling state management from authentication.

◆ SKILLS

C++	Python	Go	PyTorch	ONNX	LangChain	gRPC	Protobuf	Asio	Qt	Boost	ROS / ROS2
CMake / Bazel	GDB / Valgrind	MySQL / Redis	Linux	Git	Multithreading	Performance tuning	Transformers				
LoRA / SFT	GGUF	Docker									

Languages

C++: familiar with C++11/17, OOP, generic programming, STL and templates; solid on memory management and multithreading; familiar with Boost and streaming frameworks such as RTMP / SRT / ffmpeg

Python: familiar with Flask / FastAPI, PyTorch and OpenCV; capable of data processing and scripting

Also familiar with Go (goroutine / channel, Gin), MySQL / Redis

AI / LLM

Models & data: model post-training, data mining and dataset construction

Frameworks: PyTorch / TensorFlow, LangChain, LlamaCpp, ONNX runtime

Middleware	IO multiplexing, HTTP/HTTPS, WebSocket, MQTT, gRPC / Protobuf, ROS / ROS2, AD deployment & calibration
Tooling	Git, g++ / CMake / Bazel, GDB / Valgrind, Perf / Top / Htop — end-to-end engineering and performance tuning

◆ SUMMARY

Hands-on experience in autonomous-driving model post-training and data mining, with C++ and Python as primary languages and a solid engineering foundation. Skilled at algorithm engineering, modular refactoring and system performance optimization, familiar with core backend technologies, mainstream communication protocols and engineering tooling — able to independently develop, deploy and troubleshoot backend modules and Infra systems. Recently extended that path to LLMs: trained a small model from scratch, ran LoRA fine-tuning, quantized it to GGUF and deployed it behind a Go gateway, then measured the result with perplexity, latency and throughput.